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BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Vandalizers

Team Leader Name : Aditya Pandey

Problem Statement : #5 — AI-Powered Digital Twin of India's Climate

Team Members

Team Leader: Aditya Pandey

Name: Aditya Pandey
College: Chandigarh University

Team Member-1:

Name: Palak Rai
College: IPEC

Team Member-2:

Name: Avik Srivastava
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The Opportunity — India's climate intelligence is fragmented

How it solves the problem - the living-twin loop

- Observe — ingest real IMD daily rainfall/temperature + INSAT-3DR satellite for any day 1981→2026.
- Forecast — ClimateUNet predicts 10-day rainfall + max/min temperature in a single pass.
- Assimilate — fuse real observations into the AI state via Optimal Interpolation (keeps the twin anchored to reality).
- Detect hazards — heatwave, heavy-rain and dry-spell maps computed straight from the forecast.
- Simulate — planner pulls levers (+2 °C, greener city, cool roofs) → urban heat-stress & air-quality maps update live.



How VARUNA is Different from Existing Ideas?

Typical Weather Apps	Your System (Advanced Platform)
Just show a forecast	Reconstructs → forecasts → assimilates real observations → simulates interventions
Foreign/global models & datasets	100% Indian data (IMD + INSAT-3DR), Atmanirbhar, on-device capable
Often use synthetic/fabricated fallback data	Zero synthetic data — every field derived from real IMD/INSAT sources
Single output variable, single lead time	3 variables × 10 lead days generated in one pass
No downstream impact or actionable output	Integrated apps: hazard warning, urban heat analysis, air quality insights
Static dashboards	Live "what-if" simulator — adjust inputs and maps update instantly



List of features offered by VARUNA:

Core AI

ClimateUNet: residual U-Net + Squeeze-Excite attention, 7.42 M params, 10-day direct multi-horizon

XGBoost companion model, ensembled at city scale

Three baselines for honest benchmarking (persistence, persistence-of-anomaly, climatology)

Digital twin

Optimal-Interpolation data assimilation (RMSE-to-obs before/after)

Real INSAT-3DR satellite ingest + regrid + model cross-check

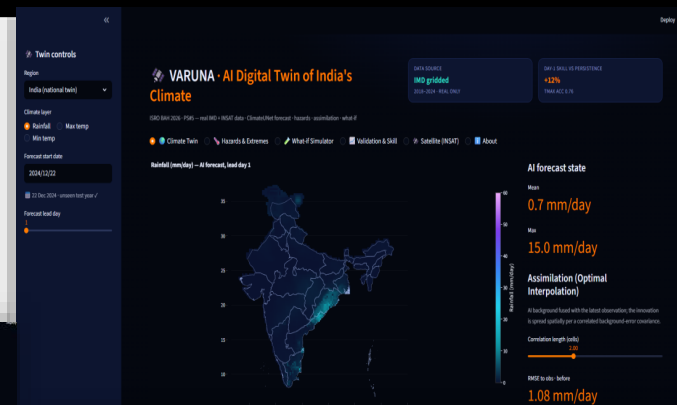
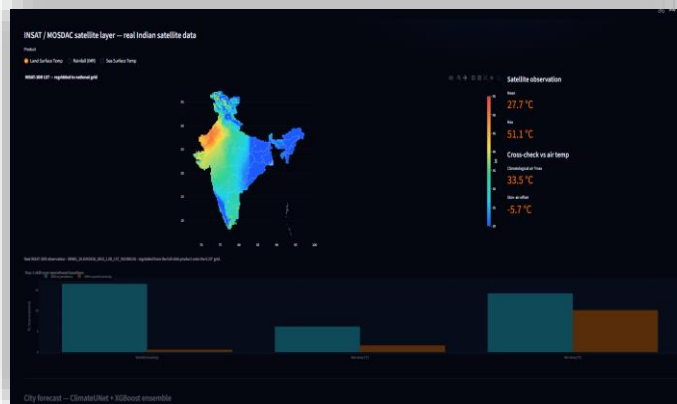
Applications

Hazard early-warning: heatwave, heavy-rain, dry-spell maps

Urban heat: NWS heat index + LST/UHI proxy

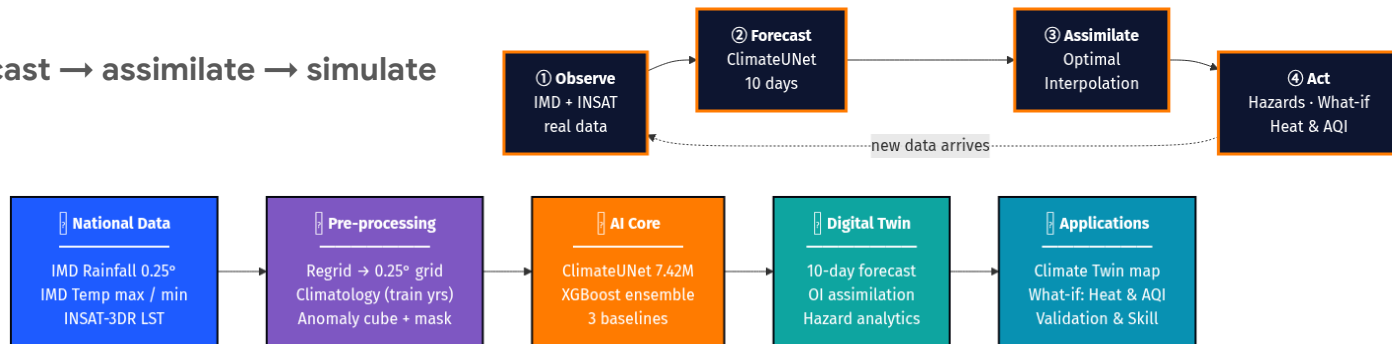
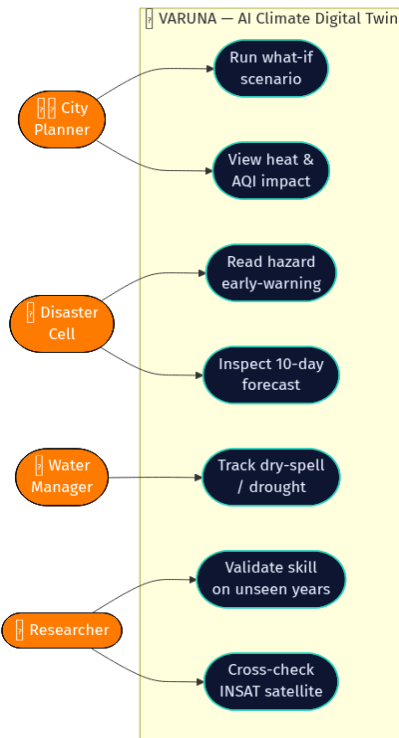
Air quality: CPCB PM2.5→AQI sub-index

What-if levers: Temperature, Rainfall, urban greening, cool-roof albedo, added built-up

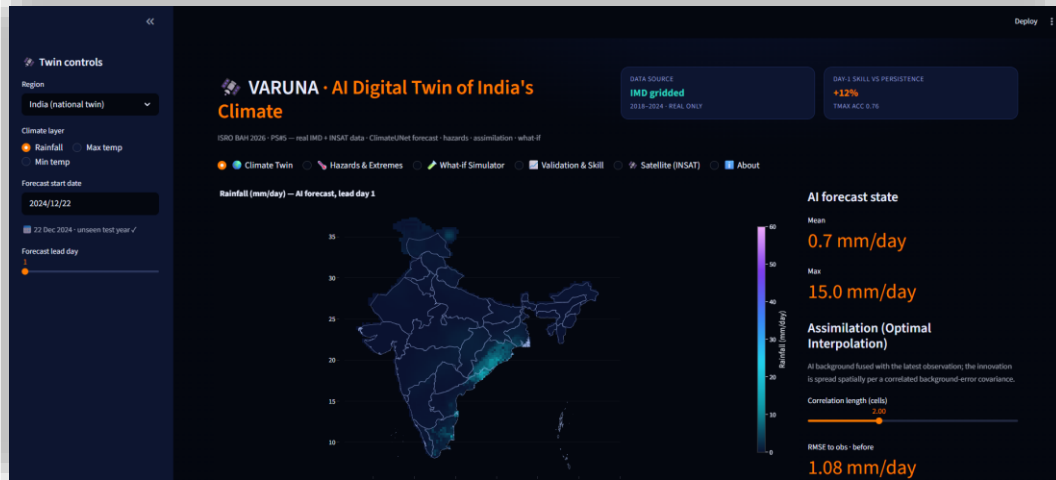


Use-case diagram

Process flow — observe → forecast → assimilate → simulate



Wireframes/Mock diagrams of VARUNA



- **City Planner** → runs what-if scenario → gets heat/AQI impact maps
- **Disaster Cell** → reads hazard layer → issues early heat/rain alert
- **Water Manager** → tracks dry-spell length → plans reservoir release
- **Researcher** → reproduces pipeline → validates skill on unseen years

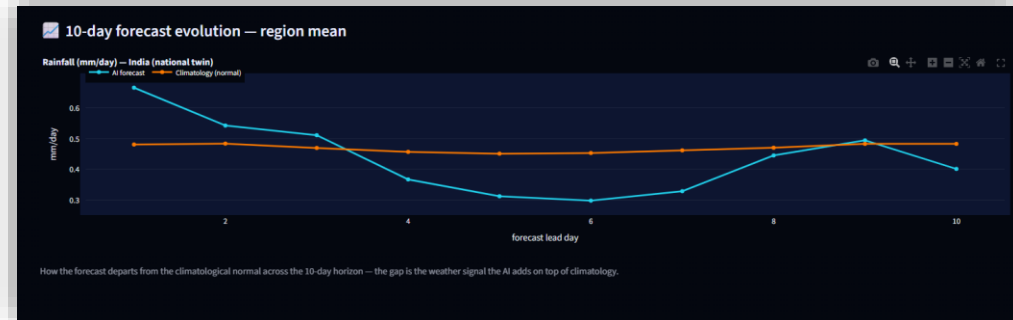
Streamlit + Plotly App

GitHub –

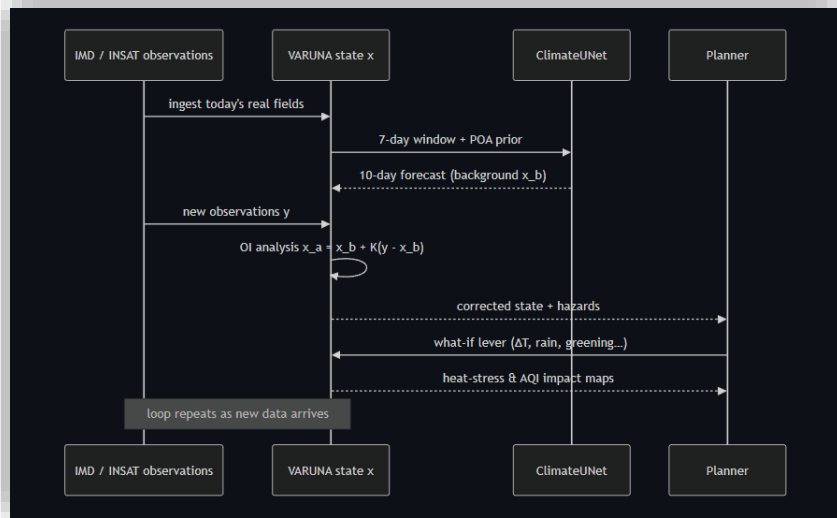
<https://github.com/Aditya060806/VARUNA>

Live Demo -

<https://huggingface.co/spaces/Aditya1002/VARUNA>

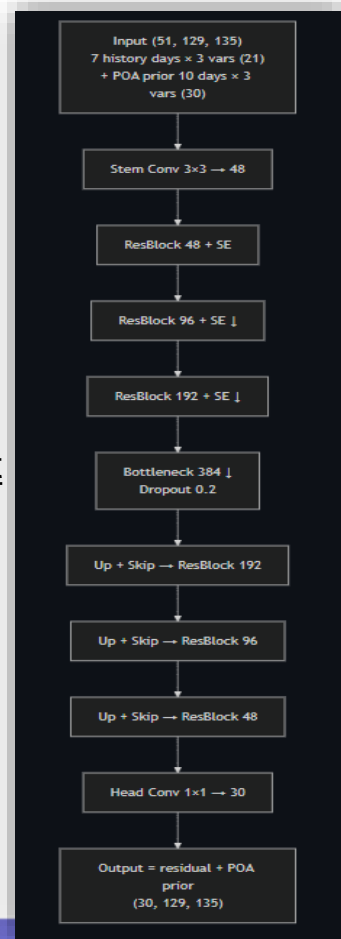


Architecture diagram of VARUNA



Digital-Twin Assimilation

The AI Model: ClimateUNet



IMD daily grids → obs cube (16071, 129, 135, 3) [rain, tmax, tmin]

→ subtract climatology, ÷ anomaly-std

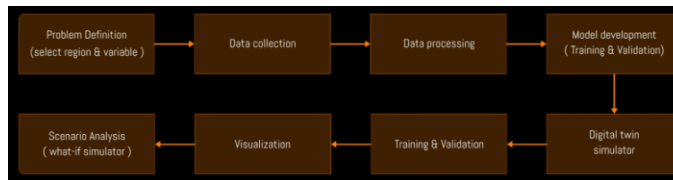
anomaly cube (16071, 129, 135, 3)

→ 7 history days + persistence-of-anomaly prior

ClimateUNet input (51,129,135) → output (30,129,135)

→ reshape + add climatology + ReLU(rain)

forecast: {rain, tmax, tmin} × 10 lead days (real units)



Technologies to be used in VARUNA:

Layer	Tools / Technologies
Data Processing	imdlb, xarray, netCDF4, h5py, bottleneck, NumPy, Pandas
Deep Learning	PyTorch 2.5, CUDA 12.1 (Mixed Precision Training)
Classical Machine Learning	XGBoost 2.0 (GPU Hist), scikit-learn
Visualization	Streamlit 1.56, Plotly, Matplotlib, Altair
Scientific Computing	SciPy (Optimal Interpolation, Regridding)
Deployment & DevOps	Docker (python:3.11-slim), Hugging Face Spaces, GitHub Actions (CI/CD)

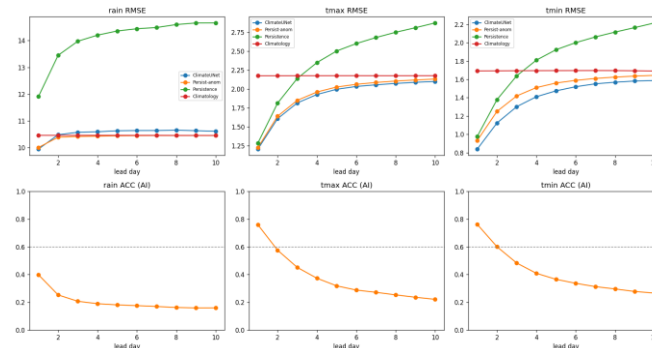
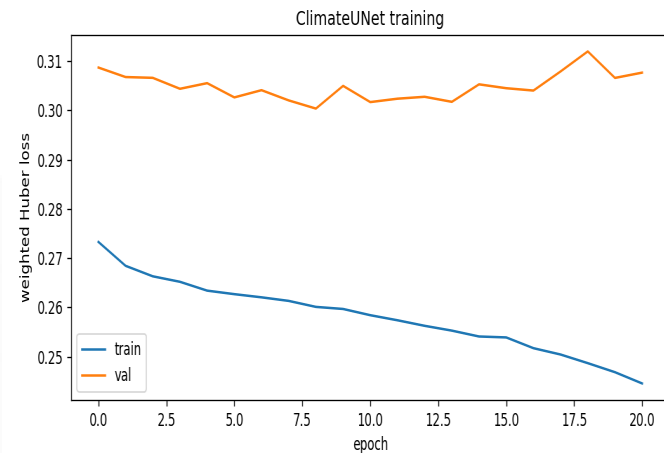
Data sources: IMD gridded rainfall 0.25° · IMD temperature 1.0° · INSAT-3DR L2B (MOSDAC).



Estimated implementation cost:

ClimateUNet — stable by design, honest by construction

Item	Prototype (Built)	Scale-up (National Operations)
Data	₹0 — Open IMD + INSAT/MOSDAC datasets	₹0 — Government-provided datasets
Compute (Training)	₹0 — Existing laptop (RTX 4050, 6 GB GPU)	₹3–6 L/year — Cloud GPU cluster for foundation model fine-tuning
Hosting	₹0 — Hugging Face Free Tier (CPU)	₹1–3 L/year — National-scale serving with tiled deployment and scheduled data ingestion
Software	₹0 — 100% Open Source	₹0 — 100% Open Source
Team	Student hackathon team	3–5 Engineers / Scientists
Total Estimated Cost	≈ ₹0 (Marginal Cost)	≈ ₹5–10 L/year (Indicative)



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THANK YOU